Customer Support Ticketing System

User Story: As a customer support agent, I want to manage and resolve customer support tickets efficiently.

Functional Expectations:

- Ticket creation by end user / customer.
- Ticket assignment, Status tracking and resolution management.
- Basic reporting on ticket resolution performance (Landing page summary for Agent).
- Self-Registration and Login Feature

Tech Stack:

Tech Stack	Backend	Frontend	Database
Java	Java 11 or Java 17, Spring Boot, Maven, JUnit, JPA-Hibernate, Logging	React / Angular	PostgreSQL
.NET Core	.NET Core 6+, C#, ASP.NET Core, Entity Framework Core, xUnit, Logging	React / Angular	PostgreSQL / MSSQL
Python	Python 3.9+, Flask / FastAPI / Django, SQLAlchemy, PyTest, Logging	React / Angular	PostgreSQL
Node.js	Node.js 18+, Express.js, Mocha/Chai, Sequelize, Logging	React / Angular	PostgreSQL

Expectations:

Technical Aspect	Expectation
Microservices Architecture	Independent, loosely coupled microservices with Synchronous RESTful APIs and asynchronous messaging where needed.
Logging	Structured logging with appropriate log levels (INFO, DEBUG, ERROR… etc);
Performance Optimization	Database optimization (indexing, query optimization), and basic load testing.
Data Validation	Strong validation on both client-side and server-side; input sanitization to prevent vulnerabilities.
Exception Handling	Graceful error handling with specific and global exception handlers and standardized error responses.
Unit Testing	Unit tests with high code coverage and mocking frameworks.
Security	Authentication (e.g., JWT), role-based access control.
Code Quality	Coding standards, Best Practices, Design Patterns, and static code analysis (e.g., SonarQube).
Configuration	Use property files, JSON config, DB config, Env Config… etc. wherever needed. Ensure to encrypt the sensitive data.